

ACADEMIC RESEARCH CASE STUDY • EMPIRICAL CODE STYLOMETRICS

Zero-Shot LLM Code Synthesis vs. Production-Hardened Human Reference Code: A Stylometric and Structural Case Study

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Abstract

Evaluating Large Language Model (LLM) code generation typically focuses on functional pass rates (e.g., HumanEval, MBPP) rather than stylometric, performance, and maintenance properties. This paper presents an empirical case study comparing **76 zero-shot synthetic code generations** produced by three frontier LLM architectures (*Google Gemini 3.5 Flash*, *OpenAI GPT-5.6 Sol*, and *Anthropic Claude Sonnet 4.6*) against a reference baseline of 10 production-hardened standard library functions authored by senior human engineers prior to the LLM era (2017–2018 code from React 16, Go 1.10, Redis 5.0, Linux Kernel 4.14, Rust stdlib, PyTorch 1.0, and FastHTTP). We also evaluate an auxiliary secondary benchmark of 50 pilot recreation generations produced by Gemini 3.5 Flash across the 10 human prompt tasks.

Quantitative analysis demonstrates statistically significant stylometric divergence: frontier synthetic implementations exhibit **+297% lines of code (LOC) expansion** (Mean = 59.62 ± 27.67 LOC vs. 15.00 ± 6.78 LOC human, Mann-Whitney U = 22.5, Holm-Bonferroni $p_{\text{adj}} = 9.04 \times 10^{-6}$, rank-biserial effect size $r_{\text{rb}} = +0.941$), elevated comment density ($13.73\% \pm 11.47\%$ synthetic vs. $1.43\% \pm 4.52\%$ human, $p_{\text{adj}} = 0.0019$, $r_{\text{rb}} = +0.686$), higher explicit type annotation density (13.13 ± 11.72 vs 1.50 ± 1.35 , $p_{\text{adj}} = 0.0004$, $r_{\text{rb}} = +0.774$), and higher vertical whitespace ratios ($17.47\% \pm 4.88\%$ synthetic vs. $5.54\% \pm 6.95\%$ human, $p_{\text{adj}} = 0.0002$, $r_{\text{rb}} = +0.812$). All confidence intervals are computed using non-parametric 10,000-resample bootstrapping.

A per-model narrow task sub-analysis reveals distinct model behavior: while GPT-5.6 Sol (32.00 ± 19.76 LOC, $p_{\text{adj}} = 0.0401$) and Gemini 3.5 Flash (34.57 ± 6.97 LOC, $p_{\text{adj}} = 0.0002$) contract significantly on narrow tasks, Claude Sonnet 4.6 remains highly verbose (63.14 ± 27.53 LOC, $p_{\text{adj}} = 0.0079$) due to embedded unit test harnesses and rich type declarations.

1. Experimental Pipeline Overview

Data Sources (Human n=10, Frontier N=76, Auxiliary N=50) —> Stylometric Parser —> Statistical Evaluation (Mann-Whitney U, Holm-Bonferroni FWER, Bootstrap 95% CIs, Kruskal-Wallis Inter-Model).

2. Introduction

Automated code generation powered by Large Language Models (LLMs) has expanded from single-line autocompletion to complete function synthesis. While LLMs achieve high pass rates on standard coding benchmarks, production software quality depends heavily on non-functional dimensions: maintainability, memory alignment, cache locality, control flow clarity, and domain safety invariants.

3. Related Work

Recent research examines LLM code quality beyond test pass rates: Binkley et al. (2023) and Jesse et al. (2023) observed significant verbosity in LLM code; Kabir et al. (2023) analyzed tutorial explanation habits in LLM outputs; Ugare et al. (2024) systematically categorized performance bugs in LLM code; and Nguyen et al. (2023) documented security and domain invariant omissions in Copilot outputs.

4. Methodology & Dataset Composition

Pre-AI Human Baseline Reference Dataset (n=10): Extracted from 2017–2018 repositories: React 16 Fiber Scheduler (Andrew Clark / Dan Abramov), React 16 shallowEqual, Go 1.10 strings.Builder (Russ Cox / Brad Fitzpatrick), Rust slice::rotate_left, PyTorch 1.0 clamp_out, Redis 5.0 raxInsert (antirez), TypeScript 3.0 isIdentifierStart (Anders Hejlsberg), Linux Kernel 4.14 eBPF htab_map_lookup_elem, FastHTTP caseInsensitiveCompare, and Rust Vec::retain.

Sampling Protocol & Rationale: Single-shot stateless generation (temperature = 0.0), max_tokens = 750-800. System prompt: "Write clean, production-quality code. Output only the code, no explanation." Gemini

Flash lower token pricing permitted 2 runs per task-language pair (N=38), while GPT-5.6 Sol (N=19) and Claude Sonnet 4.6 (N=19) ran 1 run per pair under fixed API budget constraints.

5. Quantitative Stylometric Results & Statistical Significance

5.1 Primary Frontier Model Study: Human Reference (n=10) vs. Frontier LLMs (N=76)

Stylometric Metric	Human Reference (n=10)	Frontier LLMs (N=76)	Mann-Whitney U	Raw p-val	Holm-Bonferroni p_adj	Rank-Biserial Effect Size (r_rb)	FWER Significance
Lines of Code (LOC)	15.00 ± 6.78 [11.1, 18.9]	59.62 ± 27.67 [53.5, 65.9]	22.5	p = 1.51e-06	p_adj = 9.04e-06	r_rb = +0.941 (Massive)	Significant (p < 0.01)
Comment Density (%)	1.43% ± 4.52% [0.0, 4.3]	13.73% ± 11.47% [11.2, 16.3]	119.5	p = 3.73e-04	p_adj = 0.0019	r_rb = +0.686 (Large)	Significant (p < 0.01)
Explicit Type Annotations	1.50 ± 1.35 [0.7, 2.3]	13.13 ± 11.72 [10.6, 15.8]	86.0	p = 7.47e-05	p_adj = 0.0004	r_rb = +0.774 (Large)	Significant (p < 0.01)
Helper Method Count	0.00 ± 0.00 [0.0, 0.0]	1.25 ± 2.46 [0.7, 1.8]	260.0	p = 0.0416	p_adj = 0.0832	r_rb = +0.316 (Medium)	Marginally Significant
Return Statement Count	1.70 ± 1.49 [0.9, 2.6]	3.16 ± 2.31 [2.7, 3.7]	219.0	p = 0.0272	p_adj = 0.0816	r_rb = +0.424 (Med-Large)	Marginally Significant
Vertical Whitespace (%)	5.54% ± 6.95% [1.8, 9.8]	17.47% ± 4.88% [16.4, 18.5]	71.5	p = 3.32e-05	p_adj = 0.0002	r_rb = +0.812 (Massive)	Significant (p < 0.01)

 Figure 1: Mean Lines of Code (LOC) Expansion Across Author Groups

5.2 Per-Model Breakdown & Multiple-Testing Corrected Kruskal-Wallis Significance

Model Sub-Group	Record Count (N)	Mean LOC (± SD)	Mean Comment Density (%)	Mean Type Annotations	Mean Helper Methods
Google Gemini 3.5 Flash	N=38	55.39 ± 23.81	21.31% ± 7.77%	10.97 ± 8.21	1.08 ± 1.42
OpenAI GPT-5.6 Sol	N=19	54.16 ± 35.19	1.08% ± 2.18%	10.74 ± 9.53	1.26 ± 1.85
Anthropic Claude Sonnet 4.6	N=19	73.53 ± 19.76	11.23% ± 10.61%	19.84 ± 14.21	1.58 ± 2.14
Kruskal-Wallis H-test	—	H = 8.93, p_adj = 0.0345	H = 43.50, p_adj = 1.43e-09	H = 4.98, p_adj = 0.1658	H = 0.92, p_adj = 0.6316

5.3 Per-Model Narrow Task Analysis (Testing All 3 Frontier Models)

Model	Narrow Task Count (N)	Mean Narrow LOC (± SD)	Mann-Whitney U	Raw p-val	Holm-Bonferroni p_adj	Significance vs Human Baseline
OpenAI GPT-5.6 Sol	N=7	32.00 ± 19.76	13.5	p = 0.0401	p_adj = 0.0401	Significant (p < 0.05)
Google Gemini 3.5 Flash	N=14	34.57 ± 6.97	2.0	p = 7.57e-05	p_adj = 0.0002	Significant (p < 0.01)
Anthropic Claude Sonnet 4.6	N=7	63.14 ± 27.53	5.0	p = 0.0039	p_adj = 0.0079	Significant (p < 0.01)

Empirical Finding: GPT-5.6 Sol (32.00 LOC) and Gemini 3.5 Flash (34.57 LOC) contract significantly on narrow tasks compared to their complex task outputs (70.44 ± 25.29 LOC). However, **Claude Sonnet 4.6 remains highly verbose even on narrow tasks (63.14 ± 27.53 LOC)** due to embedding executable unit test harnesses and explicit interface definitions directly within outputs.

5.4 Secondary Auxiliary Study: Human Reference (n=10) vs. Gemini Pilot Recreations (N=50)

Separately from the primary frontier model evaluations, we evaluated an auxiliary dataset of 50 pilot recreation generations produced by Google Gemini 3.5 Flash (5 independent runs per prompt for flow_01–flow_10). Synthetic outputs averaged 13.14 ± 5.21 LOC and $2.39\% \pm 6.21\%$ comment density.

A Mann-Whitney U test between the human reference functions (n=10) and the auxiliary Gemini recreations (N=50) yielded $p = 0.5176$ (LOC $U = 283.0$). A non-significant p-value ($p > 0.05$) does not prove statistical equivalence, given the modest statistical power of a small human reference sample (n=10). Rather, it indicates that no statistically significant difference was detected at the $\alpha = 0.05$ level between the human baseline and the auxiliary Gemini recreation set on narrow single-function prompts.

7. Empirical Frequency Analysis of 7 Structural Patterns

Pattern Identifier & Description	Frontier Models (N=76)	Full Synthetic Set (N=126)*	Primary % (k/76)
Pattern 1: Structural Micro-Fragmentation — Decomposing simple tasks into ≥ 2 helper functions or extra class wrappers.	68 / 76	69 / 126	89.5%
Pattern 2: Contextual Invariant Omission — Omitting domain safety checks (struct copy guards, null prototype handling, Object.is).**	0 / 76**	6 / 126	0.0%**
Pattern 3: Trivial Syntax-Echo Comments — Writing comments that directly repeat line syntax (# Increment counter).	27 / 76	28 / 126	35.5%
Pattern 4: Functional Iterator Closures in Hot Loops — Using .every(), .map(), .forEach() in performance hot loops.	4 / 76	5 / 126	5.3%
Pattern 5: In-Loop Mutating Array Shifts — Regressing runtime complexity from O(N) to O(N²) via vector removals in loops.	0 / 76	0 / 126	0.0%
Pattern 6: Asynchronous State & Timer Lifecycle Leaks — Omitting clearTimeout() or mutating subscriber lists live.***	0 / 76***	0 / 126***	0.0%***
Pattern 7: Compiler Vectorization Obstacles — Using nested std::min(std::max(...)) calls that hinder SIMD auto-vectorization.	0 / 76	1 / 126	0.0%

*Note on Full Synthetic Set: Full Synthetic Set totals include the N=50 Gemini-only auxiliary pilot set; see Section 5.4 for its distinct pilot sampling protocol.

**Note on Pattern 2: Contextual invariant omissions occurred specifically in the auxiliary Gemini benchmark recreation tasks (flow_02 shallowEqual, flow_03 Go Builder), scoring 6/50 (12.0%) in the auxiliary recreation dataset.

***Note on Pattern 6: Scored 0/126 in the quantitative dataset because Tier 2 async tasks (task_12 Event Emitter and task_14 TTL Cache) were skipped when OpenRouter API calls reached the hard payment budget limit.

8. Model Stylometric Profiles

Google Gemini 3.5 Flash (N=38): Hyper-pedagogical profile, highest task comment density (21.31% ± 7.77%), extensive JSDoc wrappers.

OpenAI GPT-5.6 Sol (N=19): Enterprise minimal profile, lowest comment density (1.08% ± 2.18%), coarse-grained global mutex locks.

Anthropic Claude Sonnet 4.6 (N=19): Comprehensive test-inclusive profile, highest task line count (73.53 ± 19.76 LOC), embedded unit tests.

9. Threats to Validity

Task Complexity & Scope Confound (Construct Validity): Comparing general research task prompts against narrow 15-line standard library routines introduces a task-scope confound. Our intra-model task analysis (Section 5.3) confirms that holding the model suite constant, complex tasks generate significantly more LOC (70.44 ± 25.29) than focused tasks (41.07 ± 21.17, p = 4.82 × 10⁻⁶). Thus, the +297% LOC bloat in general task synthesis is driven by task scope expansion and structural micro-fragmentation (89.5% rate).

Model Variance on Narrow Tasks (Model Specificity): As established in Section 5.3, Claude Sonnet 4.6 remains verbose even on narrow tasks (63.14 ± 27.53 LOC), whereas GPT-5.6 Sol (32.00 LOC) and Gemini 3.5 Flash (34.57 LOC) contract significantly.

Statistical Equivalence vs. Non-Significance (Statistical Validity): A non-significant p-value (p = 0.5176) does not prove statistical equivalence due to the limited statistical power of n=10 human reference samples.

Human Baseline Selection Bias (Construct Validity): Our human reference sample (n=10) consists of production-hardened routines from senior engineers.

Zero-Shot vs. Iterative Refinement (Internal Validity): LLMs were evaluated strictly in single-shot mode.

Sample Size & Uneven Allocation (Statistical Validity): Addressed via non-parametric Mann-Whitney U tests, Holm-Bonferroni FWER corrections, Kruskal-Wallis inter-model tests, bootstrap 95% CIs, and rank-biserial effect sizes.

10. Conclusion & References

This study establishes statistically significant stylometric differences between zero-shot LLM code and production-hardened human reference code, proving that LOC bloat is prompt-scope dependent and model-specific.

References: Binkley et al. (2023) TSE; Jesse et al. (2023) ISSTA; Kabir et al. (2023) EMSE; Nguyen et al. (2023) ICSE; Ugare et al. (2024) PACMPL.

Appendix A: Task Catalog & Reference Mapping Table

Task / Flow ID	Category	Language	Title / Task Prompt	Human Reference Source
flow_01	Benchmark Flow	JavaScript	React 16 Fiber Scheduler Min-Heap (push/siftUp)	Facebook React 16 (A. Clark / D. Abramov)
flow_02	Benchmark Flow	JavaScript	React 16 Shallow Property Comparator (shallowEqual)	Facebook React 16 (D. Abramov)
flow_03	Benchmark Flow	Go	Go 1.10 strings.Builder.WriteString	Go Standard Library (R. Cox / B. Fitzpatrick)
flow_04	Benchmark Flow	Rust	Rust slice::rotate_left	Rust Standard Library Core Team
flow_05	Benchmark Flow	C++	PyTorch 1.0 clamp_out Tensor Operator	PyTorch Core (A. Paszke / S. Chintala)
flow_06	Benchmark Flow	C	Redis 5.0 Radix Tree Insert (raxInsert)	Redis Core (S. Sanfilippo - antirez)
flow_07	Benchmark Flow	TypeScript	TypeScript 3.0 Scanner (isIdentifierStart)	Microsoft TypeScript (A. Hejlsberg)
flow_08	Benchmark Flow	C	Linux Kernel 4.14 eBPF Hashtable (htab_map_lookup_elem)	Linux Kernel (A. Starovoitov / D. Borkmann)
flow_09	Benchmark Flow	Go	FastHTTP caseInsensitiveCompare	FastHTTP Engine (A. Valialkin)
flow_10	Benchmark Flow	Rust	Rust Vec::retain In-Place Predicate Filtering	Rust Standard Library Core Team
task_01 .. task_14	Research Task	Py / JS	14 Algorithmic Tasks (LRU, Trie, Dijkstra, TokenBucket, etc.)	Benchmark Algorithmic Problems